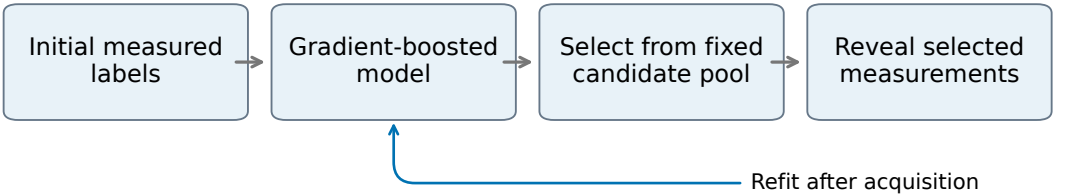
a Retrospective selection and learning



Reported permeability runs use source standardisation from all source labels. These are transductive estimates; prospective bias remains unquantified.

b Distinct questions require distinct controls

Acquisition benefit	Generalisation	Benchmark validity
Loop versus frozen fit and matched random	Held-out ranking; source-cluster uncertainty	No-target controls; null-calibrated gate